New Course Syllabus- Format (Aug-Dec 2026)

Course Name: Agentic AI

Course Code: (To be assigned)

Name of the Faculty: Dr. Sunil Saumya

Credit: L-T-P-S-C (3-0-0-4-4)

Required knowledge exposure	- Programming proficiency in Python - Familiarity with machine learning and NLP concepts is helpful but not mandatory - Knowledge of Transformer Architecture is helpful but not mandatory
Course objective	This course introduces the principles, architectures, and implementation of intelligent AI agents powered by Large Language Models. Students will learn to design autonomous systems capable of reasoning, planning, memory management, tool usage, retrieval, and multi-agent collaboration using modern frameworks such as LangGraph, LangChain, CrewAI, and Model Context Protocol (MCP). The course emphasizes building production-ready agentic applications while addressing evaluation, safety, and deployment considerations.
Course Learning outcome	Upon successful completion of this course, students will be able to: **LO1:** Explain the principles, architectures, and workflows of Agentic AI systems. **LO2:** Design and implement autonomous AI agents using LangGraph and related orchestration frameworks. **LO3:** Develop intelligent agents capable of reasoning, planning, memory management, tool usage, and Retrieval-Augmented Generation (RAG). **LO4:** Build collaborative multi-agent systems using contemporary agent frameworks and communication protocols. **LO5:** Evaluate, deploy, and monitor production-ready AI agents while incorporating safety, guardrails, and ethical AI practices.
Course Syllabus	**Unit I: Foundations of Agentic AI** Evolution of Artificial Intelligence from traditional AI to Machine Learning, Deep Learning, Generative AI, and Agentic AI; characteristics, architecture, and components of AI agents; Large Language Models (LLMs) as reasoning engines; prompt engineering techniques for agentic applications; agent design patterns; overview of

	modern agent development frameworks including LangChain, LangGraph, CrewAI, AutoGen, and Semantic Kernel; applications of Agentic AI across various domains; ethical, responsible, and trustworthy AI agent design. **Practice**: Designing prompts for agentic tasks and developing simple agent workflows using LangGraph. **Unit II: Agent Workflows and Orchestration** LangGraph fundamentals including state, nodes, edges, graph execution, and workflow orchestration; sequential, conditional, parallel, and iterative workflows; reasoning and planning strategies including ReAct, Plan-and-Execute, and Reflection; tool calling, function calling, and structured outputs; memory management techniques including short-term, long-term, and persistent memory; human-in-the-loop workflows for interactive decision-making. **Practice**: Building tool-enabled AI agents, memory-aware conversational agents, and intelligent personal assistants using LangGraph. **Unit III: Advanced Agentic AI Systems** Retrieval-Augmented Generation (RAG) and Agentic RAG; knowledge graph-enhanced agents; multi-agent systems including supervisor-worker, planner-executor, debate, and collaborative agent architectures; Model Context Protocol (MCP); context engineering; autonomous task planning; long-horizon agents; computer-use agents and web-browsing agents; design and implementation of collaborative AI systems for complex problem-solving. **Practice**: Developing Agentic RAG applications, implementing multi-agent collaboration using CrewAI or AutoGen, and building MCP-enabled agent workflows. **Unit IV: Production Agentic AI** Evaluation and benchmarking of AI agents; observability and tracing using LangSmith; guardrails, structured outputs, and safety mechanisms; responsible AI, security, and tool permission management; cost optimization strategies for LLM-powered agents; deployment of AI agents using Streamlit and FastAPI; enterprise applications of Agentic AI; emerging trends including Agent-to-Agent (A2A) communication, AI Operating Systems, and self-improving autonomous agents. **Practice**: Deploying and monitoring production-ready AI agents and developing an end-to-end capstone project.
Books	Text Books: 1. An Illustrated Guide to AI Agents by Maarten Grootendorst, Jay Alammar, Oreilly, 2026.

	2. Alammar, Jay., Grootendorst, Maarten. Hands-On Large Language Models. Japan: O'Reilly Media, 2024. Reference Books: ● **AI Agents**: **The Illustrated Guidebook** (**2025 Edition**) by Avi Chawla and Akshay Pachar.
Assessment components	Quizzes: 10% Midterm: 20% Project: 50% Class Participation & Attendance: 20%
Attendance policy	As per institute norms
